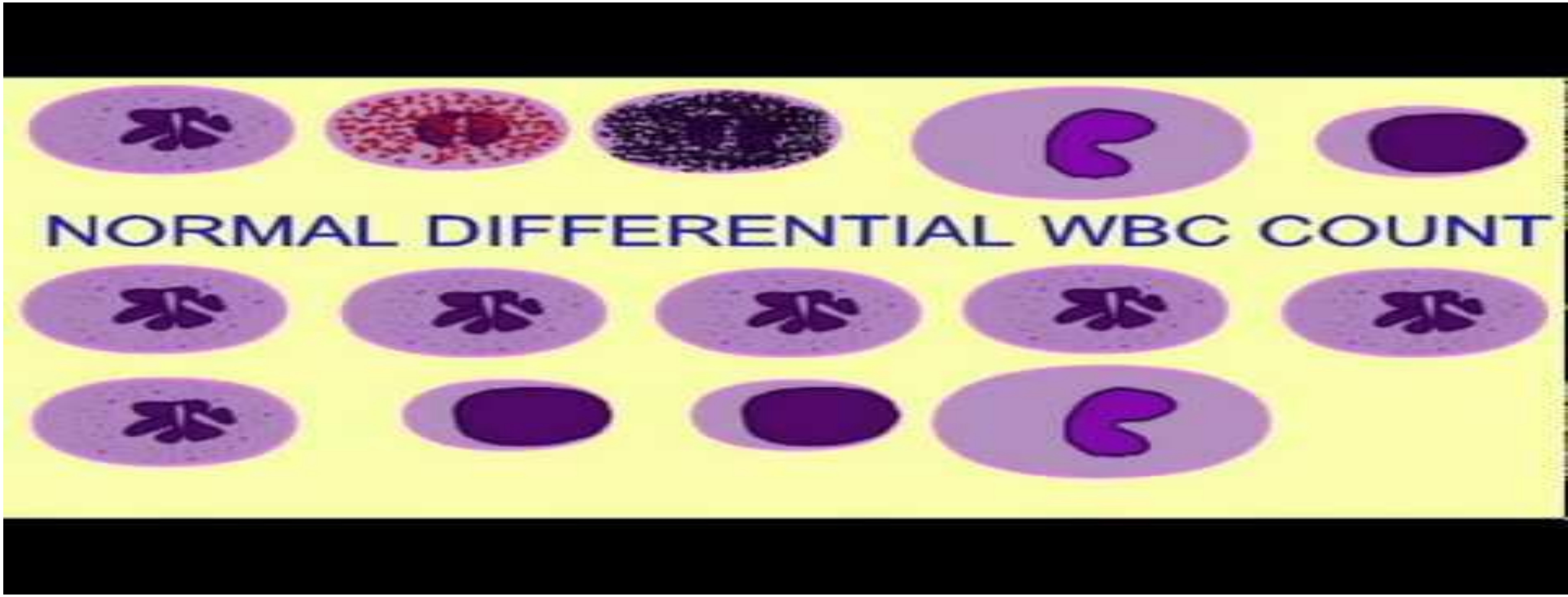
DIFFERENTIAL LEUCOCYTE COUNT



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OBJECT:

- To find out your differential leucocytes count of your own blood

REQUIREMENTS:

- Glass slides
- Leishman’s stain
- Distilled water
- Staining tray
- Glass rod
- Microscope
- Cedar wood oil and
- Lancet

THEORETICAL EXPLANATION:

- The white blood cell may be classified in to:
- **Granulocytes:** Cell having granules in their cytoplasm
- **Agranulocytes:** cells in which granules are absent
- The **granulocytes** are **Neutrophils**, **eosinophils** and the **basophils** and **agranulocytes** are **Lymphocytes** and **monocytes**

THEORETICAL EXPLANATION:

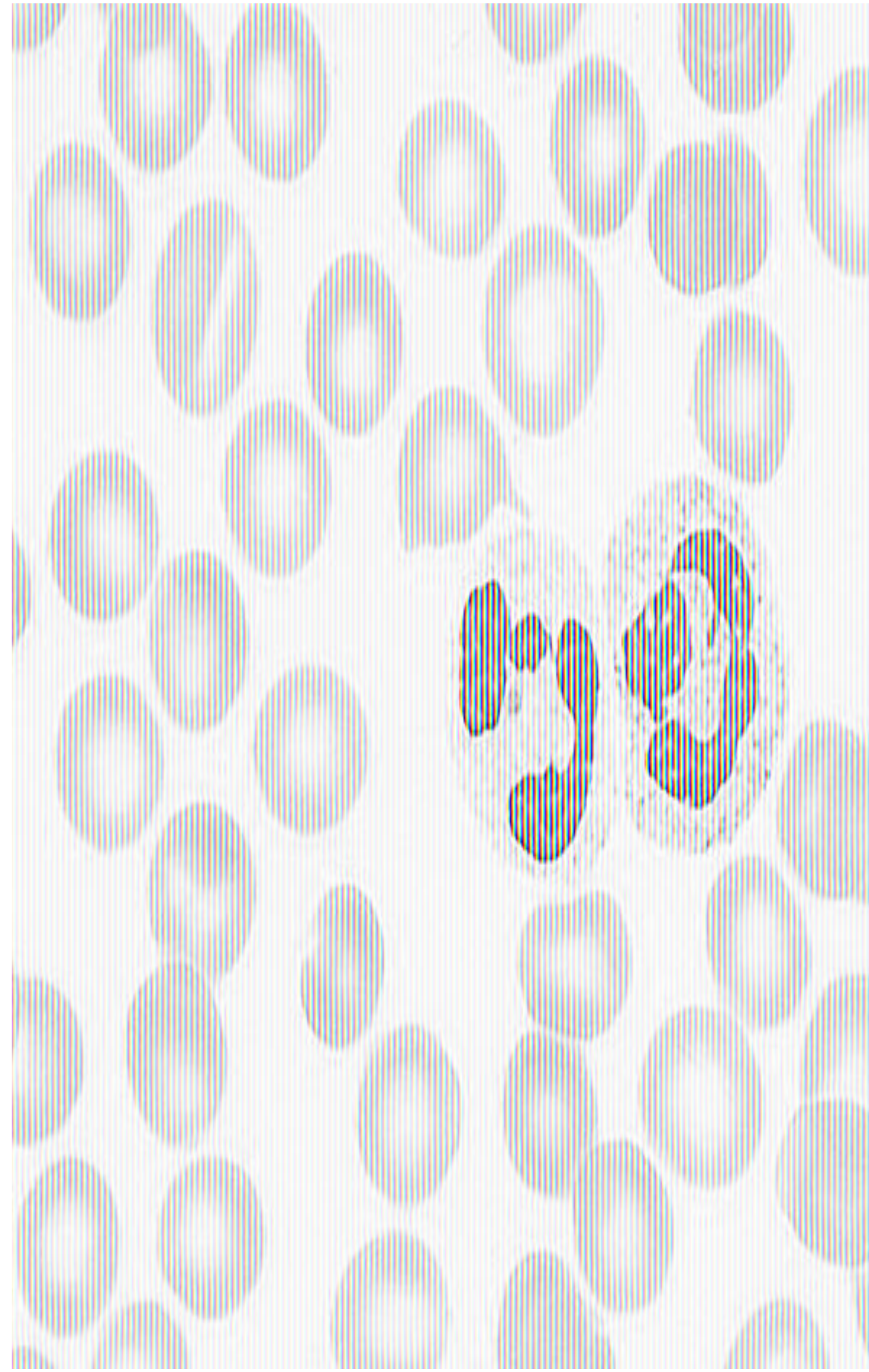
- These five types of White blood cells may be:
- Differentiated on the basis of **morphology of nucleus** of leucocytes
- Differentiated on the basis of **staining property of granules** present in cytoplasm of granulocytes. The **Leishman's stain** in the blood film. Its a mixture of two dyes
- ***Eosin:***
- This dye is **acidic** and stains the acidophilic material in cell. Its color is **red**
- ***Methylene blue:***
- This dye is **basic** and stains the basophilic material in cell
- Its color is **blue**

MORPHOLOGY OF NUCLEUS:

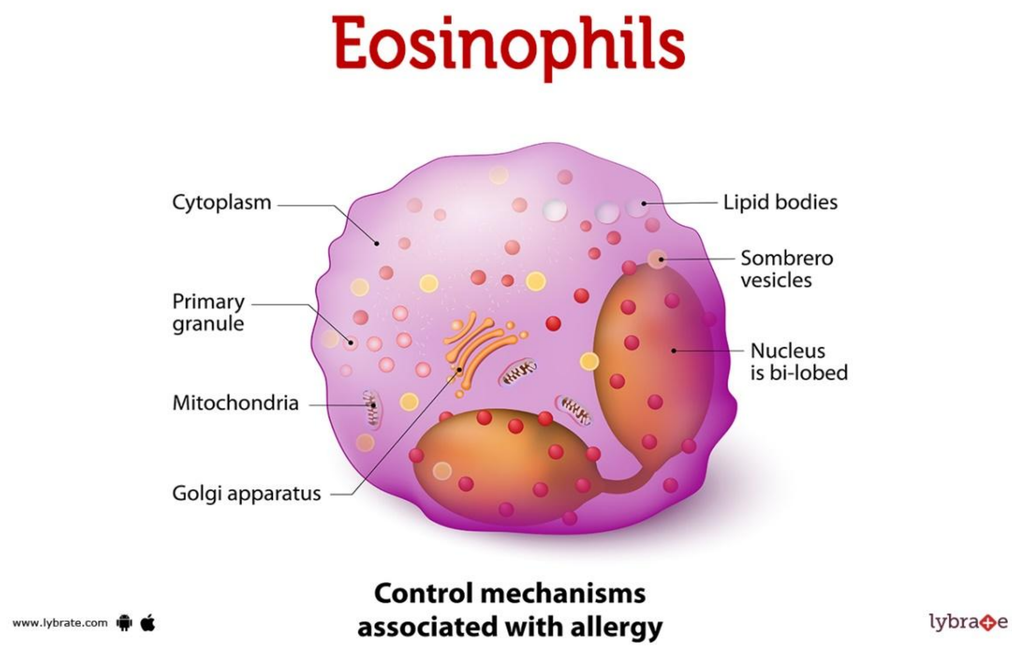
- The **nucleus** of all leucocytes is **basophilic** in nature and accepts the color. In microscope the morphology of the polymorphonuclear cells, is different.

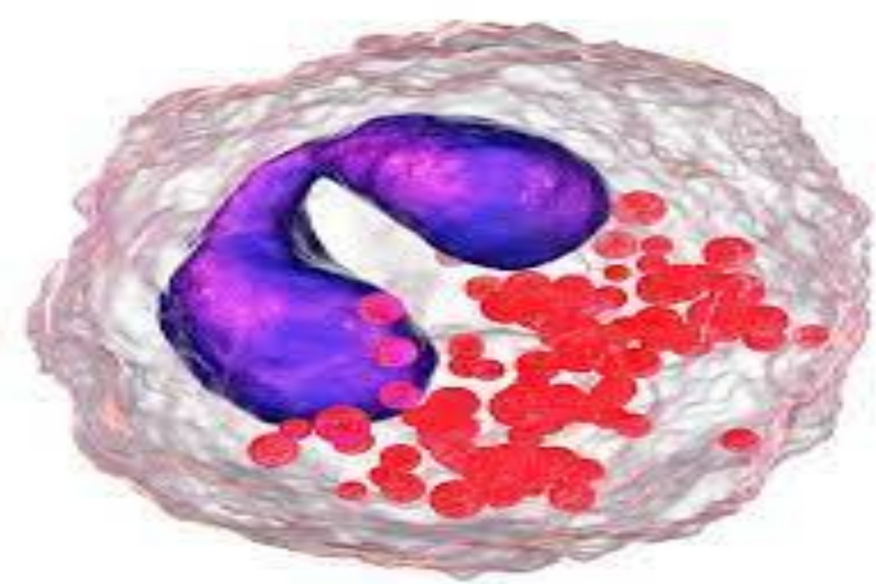
- **Neutrophils:**
- The nucleus of neutrophils is multilobed .
- The no. of lobes varies from 2-6 but mostly no .of lobes is 3 or 4.
- COUNT= 50-70%.

- Most common of the WBCs.
- Serve as the primary defense against infection.
- They increase in number in Response to infection or serious Injury.



- **Eosinophils:**
- Eosinophils have a nucleus which i either bi-lobed or S shaped
- play a role in allergic disorders .
- Count= 2-4%.
- <u>Elevations in eosinophil counts are associated with:</u>
- Allergic reactions
- Parasite infections
- Chronic skin infections
- Some cancers
- <u>Decreases in eosinophil count :</u>
- Stress
- Steroid exposure
- Anything that may suppress WBC production generally





Basophils

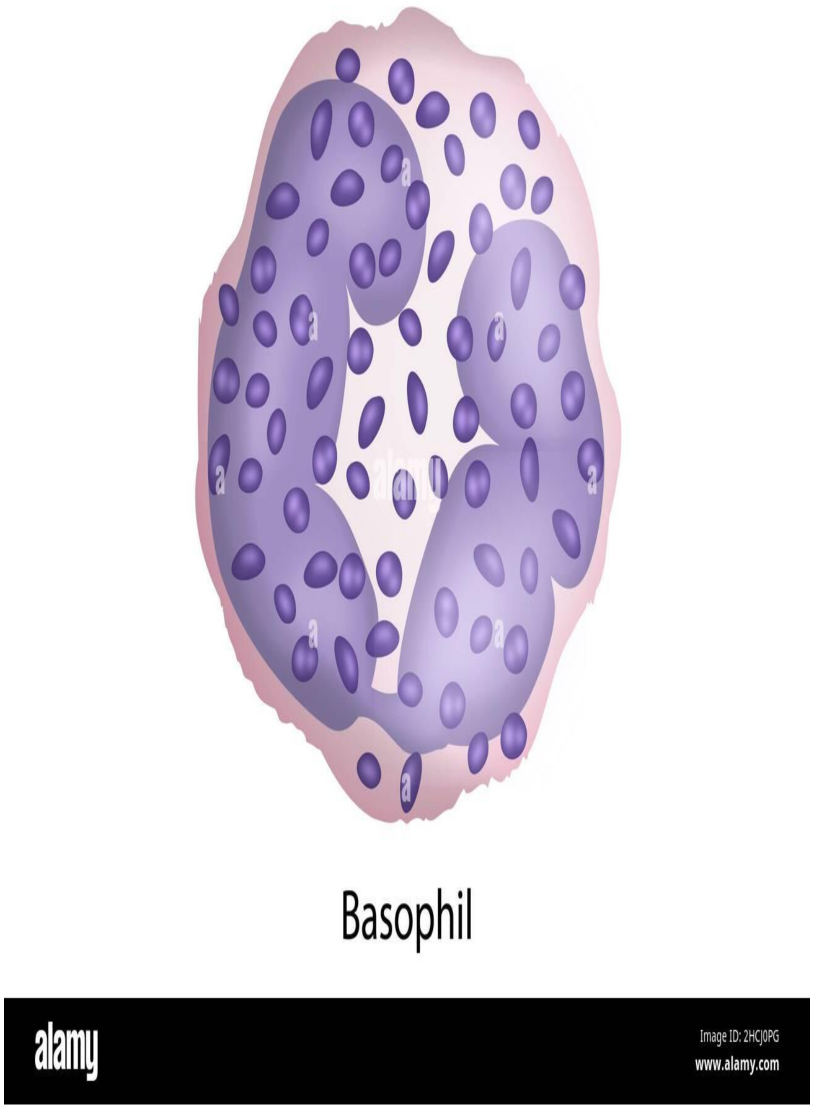
- Basophils have a nucleus which is bilobed .
- Count= < 1%.
- can digest bacteria and other foreign bodies (phagocytosis) .

 Elevations in basophil counts are associated with:

- Some cancers
- Some allergic reactions
- Some infections
- Radiation exposure

 Diminished basophil counts :

- Stress reactions
- Some allergic reactions
- Hyperthyroidism
- Prolonged steroid exposure



Lymphocytes:

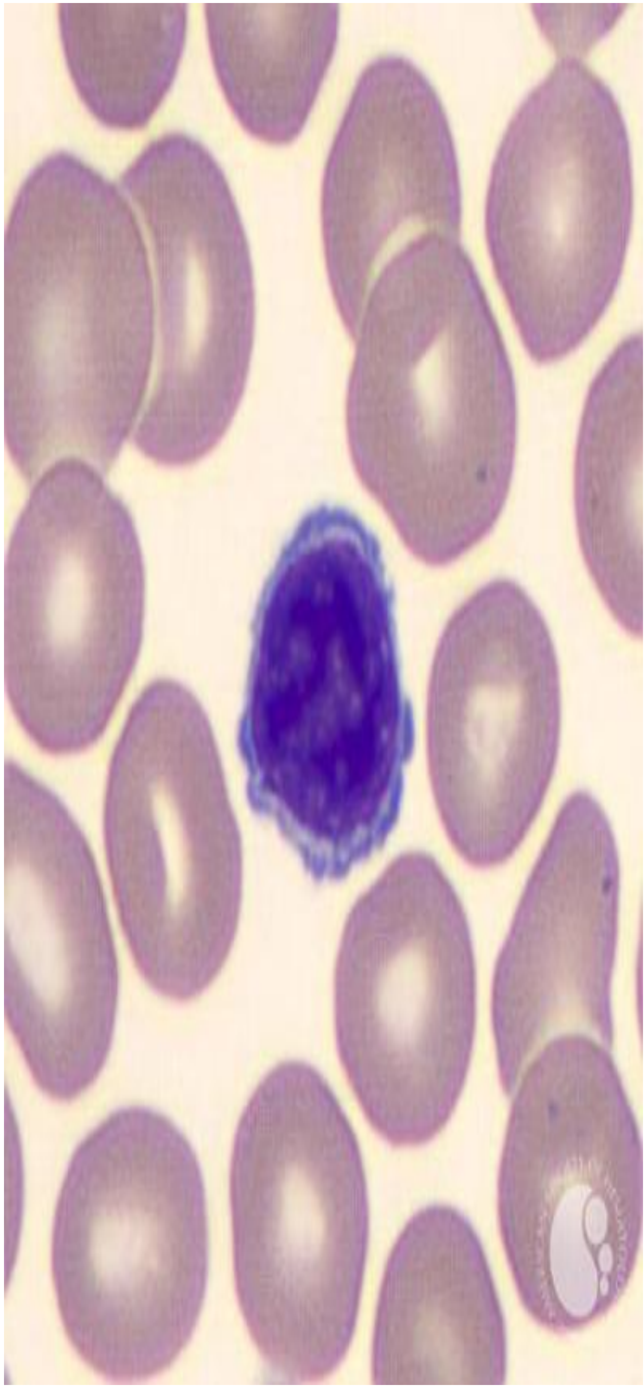
- Nucleus is mostly very large. sometimes filling all the space of the cell and only a line of cytoplasm is visible on one side of nucleus.
- Count= 20-30%.
- play both an immediate and delayed role in response to infection or inflammation.

 Increased numbers of lymphocytes are seen in:

- Most viral infections
- Some bacterial infections
- Some cancers
- Graves' disease

 Decreased numbers of lymphocytes :

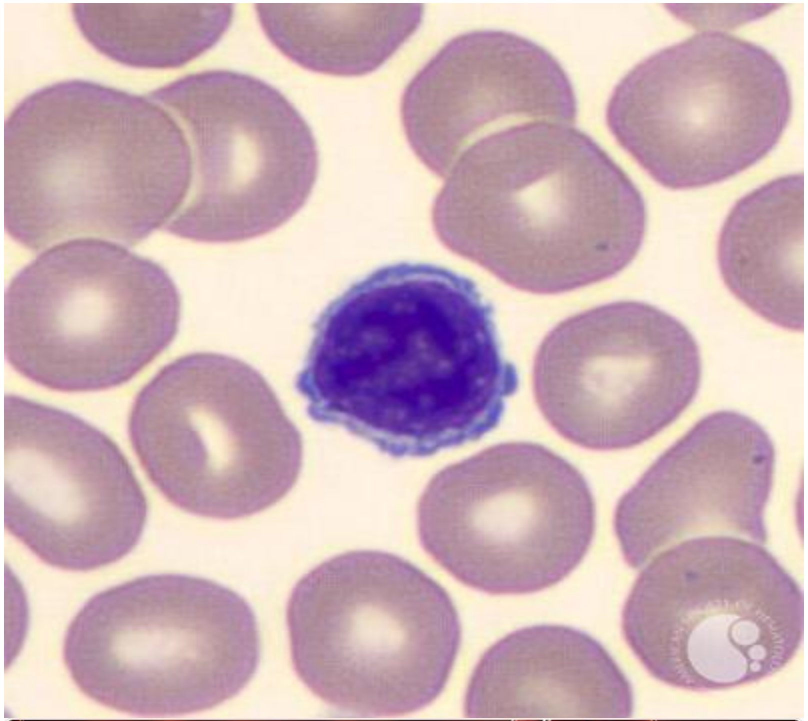
- Steroid exposure
- Immunodeficiency
- Renal failure

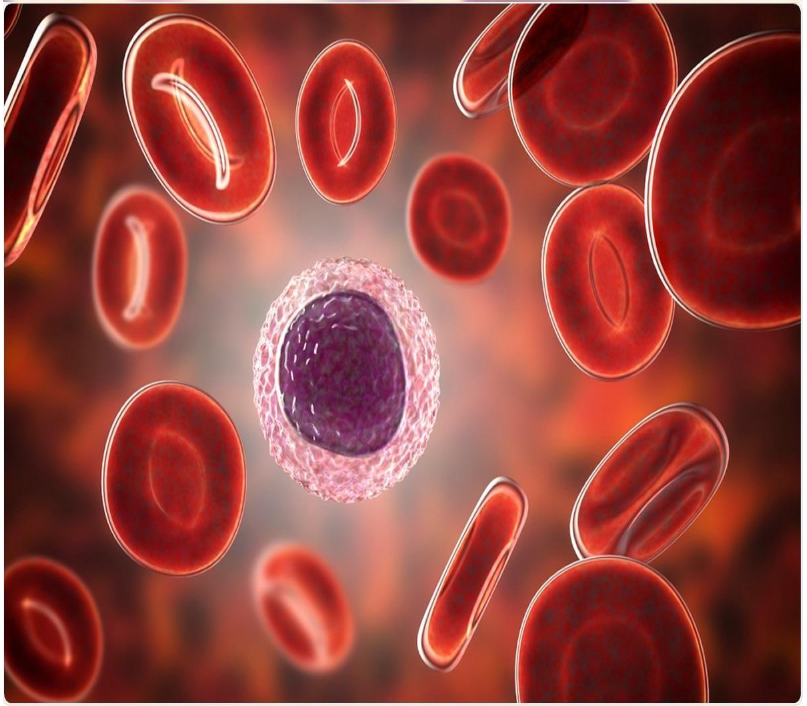


Monocyte

- monocytes have a kidney shaped nucleus or horse shoe shaped . or sometimes round making a nudge on one side of nucleus

- Count= 2-8%
- These cells respond to inflammation, infection and foreign bodies by ingesting and digesting the foreign material.
- Increased monocyte counts are associated with:
- Recovery from an acute infection
- Viral illness
- Parasitic infections
- Collagen disease
- Some cancers
- Decreased monocyte counts :
- HIV infection
- Rheumatoid arthritis
- Steroid exposure
- Some cancers





GRANULOCYTES MAY BE DIFFERENGTIATED ON THE BASIS OF STAINING PROPERTY OF GRANULES

- **EOSINOPHILS**
- Have fine and numerous granules which accept the color of eosin hence the name eosinophils i.e., eosin loving
- Appear **red** or **pink** in color

- **BASOPHILS**
- Have dense granules, and less in number than eosinophils which accept the color of the basic dye i.e., **methylene blue**, that is why they are named **basophils** i.e., basic dye loving
- Appear **blue** in colour

- **NEUTROPHILS**
- They have the granules which are neutral in character and accept the color of both **eosin** and **methylene blue** and are visible as **reddish blue granules**
- On the basis of **morphology of nucleus** and **staining property of the granules**, the stained leucocytes may be identified

CLINICAL SIGNIFICANCE:

- An increase in the WBC count i.e. leukocytosis may result by an increase in any one of the leucocyte type. These include:

1. **Neutrophilic leukocytosis** (neutrophilia) due to **appendicitis**, **rheumatic fever**, **chicken pox** and other types and **hemorrhage** or due to **physiological neutrophilia** which occurs after **exercise**

2. **Lymphocytic leukocytosis** (lymphocytosis) due to chronic infection such as **syphilis**, **pernicious anemia**, **influenza** and **hepatitis**

CLINICAL SIGNIFICANCE:

- **3. Eosinophilic leukocytosis** (eosinophilia) due to parasitic infections such as **trichomoniasis**, **psoriasis**, **bronchial asthma** and **hay fever**

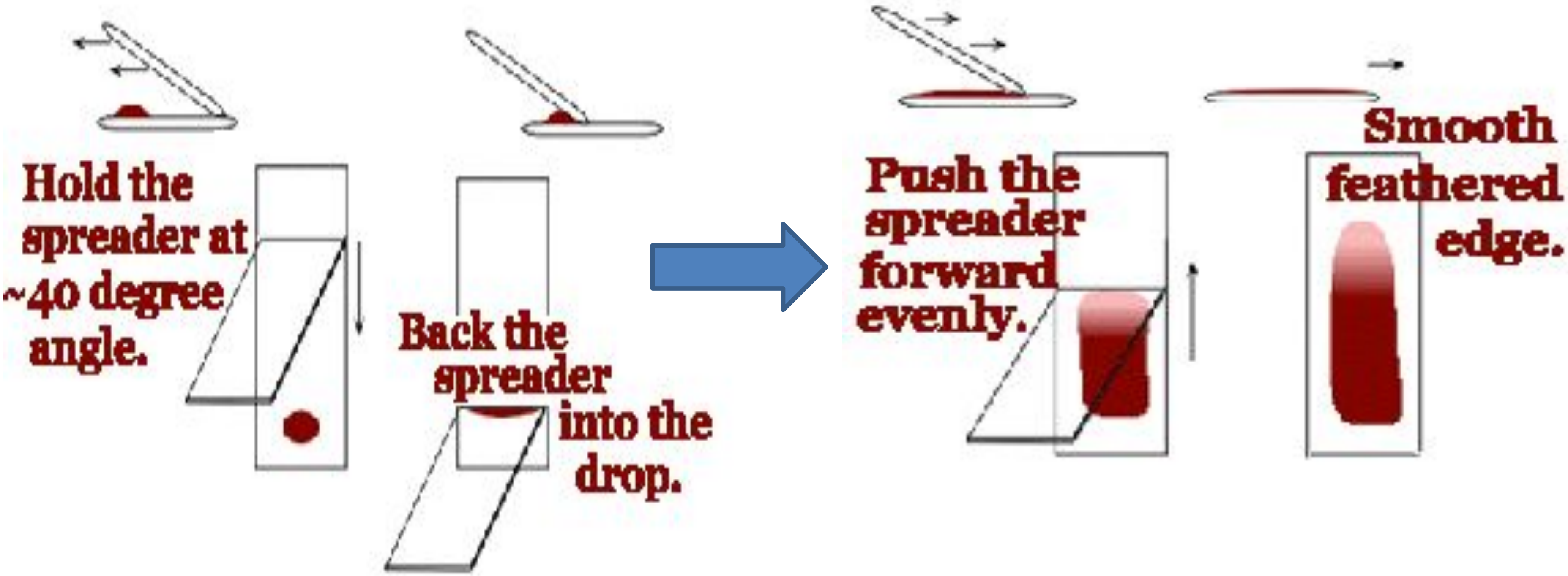
- **4. Basophil leukocytosis** (basophilia) due to **hemolytic anemia**, **chicken pox**, **measles**

- **5. Monocyte leukocytosis** occurs due to **malaria**, **bacterial endocarditis** and **typhoid fever**

- **Leucopenia** or **Agranulocytosis** results due to the decreased number of neutrophils or absence of neutrophils leaving the body unprotected and the whole body becomes infected by pathogenic bacteria

PROCEDURE:

1. Take a clean slide and put it on the table
2. Prick the finger carefully as directed previously with all the precautions
3. Take an average sized drop of blood on one side of the slide
4. Take another slide in between the thumb and index finger holding it at an angle of 45 degree in the center of the slide
5. Move the slide backwards so that it touches the drop of blood. The blood will spread along the edge of the slide
6. As the drop spreads, move the slide forward smoothly with a moderate speed so that the blood film is formed. If the edge of the spreader (slide) is rough, then the leucocytes accumulate



Hold the spreader at ~40 degree angle.
Back the spreader into the drop.
Push the spreader forward evenly.
Smooth feathered edge.

PROCEDURE:

- At the tail, a good smear has 3 qualities:

✔It is a thin film
✔It is a smooth film
✔It forms a tail

- Start the inspection slide from the head to tail

Examine at least 100 cells. For better estimation, 200 cells should be counted

DLC should not be done in badly prepared films. In bad films, lymphocytes are concentrated in the center and neutrophils shift to the periphery

In a well-prepared slide, leucocytes are uniformly distributed. Adequate number of cells can be seen in the body of the slide

PROCEDURE:

1. Place the slide in staining rack and pour the leishman’s stain so that the stain covers the entire smear
2. Stain the blood smear for 5 minutes. The stain should not be dried on the slide. If dried, pour more Leishman’s stain
3. Dilute the stain by adding slowly distilled water till the stain is washed away
4. Dry the slide in air, put one drop of cedar wood oil and focus in 100x (oil immersion lens) in microscope

PROCEDURE:

5. Identify, count and fill the observations chart

6. Start counting each type of cell from thick end of the film towards the thin end. When the counting is complete, proceed from the thin to thick part, by selecting another position

7. The observation gives the results directly. Count the types of WBC and write the results

8. The slide may be preserved by washing the cedar wood oil by xylol. The xylol swab is used to clean the oil immersion lens before using it

9. The washed slide should look **pink** in color when held in light. In a good film, the corpuscles should be spread out evenly, no rouleaux being seen. RBCs should stain **pink**

PRECAUTIONS:

- While preparing the slides, the hand or the little finger should not touch the table as it will break the smear
- The smear should be thin, smooth and form tail
- The drop of blood should not be small, as it will not form smear and it should not be so large as it will not form the tail, and neutrophil accumulate at the end of the smear
- **COUNTING:**
- Count, identify, record 100 WBCs
- Compare your result with normal percentage

OBSERVATIONS:

- **RESULT:**
- Neutrophils =
- Lymphocytes =
- Eosinophils =
- Monocytes =
- Basophils =

Relevant Questions

Q-1-On what basis will you recognize the different Leucocytes?

Leukocytes (white blood cells) are primarily differentiated based on the presence or absence of cytoplasmic granules and the shape and staining characteristics of their nucleus. Additionally, their relative size compared to red blood cells and their proportions in a blood sample are also considered.

Q-2How do your results compare with the normal range

To understand how your lab results compare with the normal range, you should always refer to the reference range provided on your specific lab report. These ranges are specific to the lab's testing methods and are designed to help interpret your individual results.

Q-3What difficulties usually arise in identification of cell type?

Identifying cell types can be challenging due to cell-to-cell variability, the existence of rare cell populations, and the dynamic nature of gene expression. Additionally, technical limitations in single-cell RNA sequencing (scRNA-seq) can introduce noise and biases that complicate accurate cell type identification.

Q-4Which are the conditions causing Eosinophilia?

The **most common causes** of a high number of eosinophils (called eosinophilia or hyper eosinophilia) are

- Allergic disorders including medication sensitivities,asthama,allergic rhinitis and atopic dermatitis
- Infections by parasites
- Certain cancers, Cancers that cause eosinophilia include Hodgkin lymphoma, leukemia, and certain myeloproliferative neoplasms

Q-5-How you can differentiate acute and chronic stages of disease on the basis of DLC?

A differential leukocyte count (DLC) can provide clues about whether a disease is in its acute or chronic stage, primarily by observing the types and proportions of white blood cells (leukocytes) present.

Acute conditions often show a prominent increase in specific white blood cells involved in the initial immune response, while chronic conditions may exhibit a more complex pattern of white blood cell changes reflecting ongoing inflammation or immune system dysregulation.